# QUANTAMED

*Quantum-Enhanced Precision Drug Discovery Platform*

## PRODUCT REQUIREMENTS DOCUMENT

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| **Target Demo Patient** | Gabi — Treatment-Resistant Epilepsy with Valproic Acid PCOS Side Effects |
| **Classification** | Confidential — Internal Use Only |

**Built with: Qiskit VQE | TRIBE v2 | Reinforcement Learning | DeepChem | RDKit | AutoDock Vina**

# 1. Executive Summary

QuantaMed is a quantum-enhanced, AI-powered precision medicine platform designed to radically compress the earliest and most expensive phase of pharmaceutical drug discovery: candidate selection and prioritization. By combining quantum variational simulation, multi-agent behavioral modeling, reinforcement learning, and pharmacogenomic profiling, QuantaMed moves the question of "which drug is right for this patient?" from a trial-and-error clinical process to a computationally grounded prediction engine.

The platform is anchored to a single, emotionally resonant case study: Gabi, a young woman with treatment-resistant epilepsy who has been prescribed valproic acid. Over several months, Gabi develops polycystic ovarian syndrome (PCOS) — a serious hormonal side effect caused by valproic acid's unintended binding to androgen-related biological pathways. The system asks: could we have predicted this? Could we have recommended lamotrigine, a safer alternative, before Gabi ever experienced harm?

QuantaMed answers yes — and demonstrates how across six interconnected simulation layers working in concert.

## 1.1 Value Proposition

**"For neurologists, pharmacists, and drug discovery researchers, QuantaMed is a quantum-hybrid simulation platform that identifies and prioritizes the safest, most effective drug candidate for a specific patient's biology — before that patient takes a single pill."**

## 1.2 The Problem We Are Solving

The global pharmaceutical pipeline is broken along three distinct fault lines:

**Fault Line 1 — Time:** Bringing a drug from discovery to market takes an average of 12 to 15 years. The earliest phase — identifying and narrowing down candidates — alone can take 3 to 5 years, largely because predictions about efficacy and safety are poor until late-stage clinical trials.

**Fault Line 2 — Money:** The average cost of developing a single approved drug exceeds $2.6 billion USD (including the cost of failures). Approximately 90% of candidates that enter Phase I clinical trials will never reach patients. The primary driver of this failure rate is inadequate early-stage prediction.

**Fault Line 3 — Personalization:** Current drug discovery treats patients as population averages. A drug approved for "depression" is prescribed to patients who may have wildly

different metabolic profiles, genetic variants, and co-medication interactions. The same antidepressant that works for one patient causes cardiac arrhythmia in another — not because the drug is bad, but because the patient's CYP2D6 enzyme makes them a poor metabolizer and the drug accumulates to toxic levels. Nobody checked.

QuantaMed directly attacks all three fault lines: it compresses candidate selection using quantum simulation (speed), it eliminates poor candidates before synthesis (cost), and it personalizes predictions using pharmacogenomic profiling (personalization).

## 1.3 What Makes QuantaMed Unique

At least a dozen drug-drug interaction (DDI) checkers exist in the market today — Drugs.com, Medscape Interactions, Epocrates, and others. None of them have the following capabilities that QuantaMed introduces simultaneously:

- Quantum-level binding energy estimation using VQE on peptide fragments of target proteins
- Patient-specific pharmacogenomic risk modifiers (CYP2D6, CYP3A4, CYP2C19 status)
- Dose-adjusted DDI severity using one-compartment pharmacokinetic modeling
- Off-target protein panel scanning to predict side effects before they occur
- Blood-brain barrier penetration scoring for CNS drug candidates
- Lipinski Rule-of-Five manufacturability scoring integrated into candidate ranking
- TRIBE v2 multi-agent behavioral simulation for patient state visualization pre- and post-treatment
- Reinforcement learning treatment decision engine that balances efficacy vs. risk dynamically
- End-to-end personalized output: one ranked candidate list tailored to one patient's complete biological profile

# 2. Problem Deep Dive & Scientific Background

## 2.1 Why Protein Folding and Binding Energy Matter

Every drug works by interacting with a protein in the body. Proteins are the molecular machines that regulate every biological function — cell signaling, neurotransmitter regulation, immune response, hormone production. A drug molecule is essentially a key designed to fit into the "lock" of a specific protein — its binding pocket — and either activate or inhibit that protein's function.

The challenge is this: a protein's functional shape is determined by how it folds from a linear chain of amino acids into a three-dimensional structure. Predicting that fold is one of the hardest problems in computational biology. And predicting how a drug molecule will interact with a folded protein — its binding energy — is even harder, because the interaction is governed by quantum mechanical forces: van der Waals interactions, hydrogen bonding, electrostatic coupling, and quantum tunneling of protons.

Classical computers simulate these interactions by approximating quantum behavior. The approximation error grows exponentially with the number of electrons in the system. For a molecule with N electrons, there are 2^N quantum states to consider. A 50-atom molecule has roughly 200 electrons — meaning 2^200 states. This number exceeds the number of atoms in the observable universe. Classical computers cannot handle this; they rely on density functional theory (DFT) approximations that sacrifice accuracy for tractability.

Quantum computers don't approximate. They ARE quantum — they can represent and manipulate the exponential state space naturally using superposition and entanglement. The Variational Quantum Eigensolver (VQE) algorithm exploits this to find the minimum energy ground state of a molecular system — the true, accurate binding energy — without exponential classical approximation.

## 2.2 Why Existing DDI Checkers Are Insufficient

The current generation of drug-drug interaction checkers fails patients in five documented ways:

**Failure 1 — No Dose Awareness:** Every DDI checker flags an interaction as HIGH, MODERATE, or LOW regardless of dose. In pharmacokinetics, dose is everything. Valproic acid at 1500mg/day produces serum concentrations that saturate liver enzymes, dramatically raising the interaction risk with other hepatically metabolized drugs. At 250mg/day, the risk profile is entirely different. Existing apps ignore this completely.

**Failure 2 — No Patient Specificity:** A CYP2D6 poor metabolizer will accumulate certain drugs to 5–10× the normal plasma concentration. A patient with hepatic impairment will metabolize

drugs entirely differently. Existing apps treat all patients identically — as if "the patient" is an average 70kg adult male with normal organ function.

**Failure 3 — No Temporal Dynamics:** Some drug interactions are not immediate — they accumulate over weeks or months. Valproic acid's association with PCOS develops over 3–6 months of chronic exposure, as androgen-stimulating effects compound. No existing app has a temporal dimension — they all treat interactions as static, immediate events.

**Failure 4 — No Mechanistic Transparency:** When an app flags "HIGH interaction risk," it doesn't explain the mechanism. Is it a CYP enzyme inhibition? An additive cardiac effect? A displacement of protein-bound drug? Clinicians need mechanism to make intelligent decisions. Existing apps provide none.

**Failure 5 — No Manufacturability Signal:** From a drug discovery perspective, a candidate drug that's highly effective but cannot be synthesized at scale, or cannot be administered orally, is useless. Existing tools never consider molecular properties that govern real-world manufacturability.

## 2.3 The Pharmacogenomics Gap

Pharmacogenomics is the study of how genetic variation affects drug response. The science is well-established. The CYP450 enzyme family — particularly CYP2D6, CYP3A4, CYP2C19, and CYP2C9 — metabolizes approximately 75% of all marketed drugs. Genetic polymorphisms in these genes create four phenotypes: poor metabolizers (PM), intermediate metabolizers (IM), normal metabolizers (NM), and ultrarapid metabolizers (UM).

Poor metabolizers cannot break down certain drugs efficiently, leading to drug accumulation and toxicity. Ultrarapid metabolizers break drugs down too quickly for them to be effective. These phenotypes can mean the difference between therapeutic benefit and life-threatening adverse reaction from the exact same dose of the exact same drug.

Genetic testing for CYP status is available (tests like GeneSight and Genomind are FDA-cleared), yet the vast majority of drug prescribing decisions are made without any knowledge of a patient's CYP phenotype. QuantaMed integrates CYP status as a first-class input into every drug ranking decision.

# 3. Anchor Case Study — Gabi

**This case study is the emotional and scientific core of every QuantaMed demonstration. All system outputs will be calibrated to Gabi's profile.**

## 3.1 Patient Profile

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|---|---|
| **Patient Name** | Gabi (anonymized demo patient) |
| **Age** | 24 years old |
| **Gender** | Female |
| **Primary Diagnosis** | Treatment-Resistant Epilepsy (Juvenile Myoclonic Epilepsy subtype) |
| **Current Medication** | Valproic Acid (Depakote) — 1000mg/day, ongoing for 9 months |
| **Emerging Side Effect** | Polycystic Ovarian Syndrome (PCOS) — confirmed 6 months post-initiation |
| **CYP2D6 Status** | Poor Metabolizer (PM) — confirmed via genetic panel |
| **CYP3A4 Status** | Normal Metabolizer (NM) |
| **CYP2C9 Status** | Intermediate Metabolizer (IM) |
| **Hepatic Function** | Mildly elevated ALT (42 U/L) — borderline hepatic stress |
| **Renal Function** | Normal (eGFR 92 mL/min/1.73m²) |
| **Co-medications** | None currently (previously tried levetiracetam — discontinued due to mood effects) |
| **Weight Trend** | +7.2 kg over 9 months on valproic acid (predicted by RL model) |
| **Seizure Control** | Partial — breakthrough seizures remain 1–2× per month |
| **Patient Goal** | Complete seizure control with minimal hormonal and metabolic side effects |

## 3.2 What Went Wrong — The Biological Mechanism

Valproic acid is a broad-spectrum antiepileptic drug that works primarily by inhibiting voltage-gated sodium channels and enhancing GABA neurotransmission. However, it also has a secondary and clinically significant interaction with androgenic signaling pathways. Valproic acid increases the secretion of insulin and insulin-like growth factor (IGF-1), which stimulates androgen production in the ovaries. Elevated androgens in women cause follicular arrest, menstrual irregularity, and the characteristic polycystic ovarian morphology of PCOS.

This hormonal disruption occurs through a combination of direct hepatic enzyme induction (affecting androgen-metabolizing enzymes), weight gain (which increases peripheral androgen conversion), and direct ovarian stimulation. The effect is dose-dependent and time-dependent — it typically manifests after 3 to 6 months of sustained use, not acutely.

Critically: none of this was flagged when Gabi was prescribed valproic acid. No system checked her sex, her hormonal profile, her weight trajectory over time, or the temporal accumulation of risk. QuantaMed would have flagged all of it at the moment of prescription.

## 3.3 What QuantaMed Would Have Done

Running Gabi's complete biological profile through QuantaMed at the time of initial prescription would have produced the following outputs:

1. Quantum binding simulation of valproic acid against sodium channel target protein — result: good efficacy signal (justifies use)
2. Off-target binding panel scan — result: moderate predicted binding to androgen receptor complex and liver CYP2C9 enzyme. FLAG: "Hormonal pathway interference predicted. Elevated PCOS risk for female patients under 30."
3. Dose-adjusted DDI severity — result: at 1000mg/day, serum valproic acid concentration predicted at 78 µg/mL, within therapeutic range but at upper bound for hepatic tolerance given Gabi's IM CYP2C9 status.
4. ADMET prediction — result: Valproic acid has poor selectivity index. Predicted weight gain +6-8 kg over 12 months (FAERS population benchmark: 2.1× higher in females under 30 vs. population average).
5. Lamotrigine comparison — result: equivalent seizure control efficacy for JME, significantly cleaner off-target profile, no androgenic pathway binding, lower weight gain prediction. RECOMMENDATION: Lamotrigine preferred for this patient profile.
6. TRIBE v2 simulation — result: Gabi's agent in "valproic acid state" shows progressive reward pathway dysregulation correlated with weight gain and hormonal disruption. Gabi's agent in "lamotrigine state" shows stable reward pathway and normalized social engagement.

# 4. System Architecture — The Six-Layer Simulation Stack

QuantaMed is architected as six distinct but interconnected simulation layers. Each layer answers a specific scientific question, produces a structured output, and feeds into the next layer. The architecture is designed so that every component has a clear scientific reason for existing — nothing is decorative, and the quantum component drives real downstream decisions.

| Layer | Name | Core Technology | Primary Question Answered | Output |
|---|---|---|---|---|
| L1 | Patient Modeling | TRIBE v2 + Patient Profile Engine | Who is this patient and what is their behavioral/biological context? | Patient Agent + Genomic Profile |
| L2 | Quantum Binding Simulation | Qiskit Aer + VQE Algorithm | How strongly does this drug molecule bind to the target protein? | Binding Energy Score per candidate |
| L3 | Off-Target Safety Screening | DeepChem + ChEMBL Database | What other proteins does this drug accidentally bind to? | Off-Target Risk Matrix |
| L4 | Personalized Compatibility | CYP Phenotype Engine + PK Modeling | How does this patient's biology process this drug? | Patient-Adjusted Risk Score |
| L5 | RL Treatment Decision Engine | Rule-Based RL Environment | Should we continue, switch, or adjust this treatment? | Treatment Action Recommendation |
| L6 | Visualization & Reporting | TRIBE v2 + Plotly + ReportLab | What does the result look like for the clinician and patient? | Dashboard + PDF Report |

## 4.1 Data Flow Diagram (Narrative)

The system begins when a clinician or researcher inputs a patient profile. This profile includes demographic data, diagnosis code, current medications, co-morbidities, and — critically — CYP phenotype status from a genetic panel. This data populates a patient agent in TRIBE v2 and simultaneously initializes the Pharmacogenomic Profile object in the Python backend.

The system then loads the candidate drug library for the specified diagnosis. For Gabi's case, this library includes valproic acid, lamotrigine, levetiracetam, topiramate, and zonisamide — the five most commonly prescribed antiepileptics for juvenile myoclonic epilepsy. For each candidate, the system retrieves the canonical SMILES molecular string from PubChem.

Each candidate drug is then passed through Layer 2 (quantum binding simulation) against the target protein. For epilepsy, this is the voltage-gated sodium channel Nav1.2 alpha subunit — a 5-10 amino acid peptide fragment of the binding domain is used for computational tractability in the VQE simulation. The output is a normalized binding energy score between 0 and 1.

Simultaneously, Layer 3 passes each candidate through the off-target panel — a curated list of 8 clinically significant off-target proteins: the cardiac hERG potassium channel (QT prolongation risk), androgen receptor (hormonal disruption), five CYP450 isoforms, and the liver enzyme ALT. Any candidate with predicted off-target binding above the threshold receives a safety penalty.

Layer 4 then applies patient-specific adjustments. Gabi's CYP2C9 intermediate metabolizer status means valproic acid accumulates approximately 1.4× faster than normal. The dose-adjusted PK model predicts her actual serum concentration and adjusts the DDI severity accordingly.

Layer 5 runs the RL environment. The state vector encodes current seizure frequency, side effect burden, adherence estimate, and time on therapy. The RL policy — trained on synthetic rule-based reward signals — evaluates whether to continue the current therapy or switch to the top-ranked alternative.

Layer 6 renders all outputs: the drug comparison matrix, the TRIBE v2 before/after brain state visualization, the risk timeline graph, and the exportable PDF report formatted for clinical use.

# 5. Detailed Feature Requirements

## F-001 — Quantum Binding Simulation Engine

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|---|---|
| **Feature ID** | F-001 |
| **Priority** | P0 — Critical Core Feature |
| **Layer** | Layer 2 — Quantum Simulation |
| **Technology** | Qiskit, Qiskit-Nature, Qiskit Aer (local simulator), VQE Algorithm |
| **Framing** | "Quantum-Ready Module" — simulated on classical hardware, designed for real QPU deployment |

### 5.1.1 Scientific Description

The quantum binding simulation engine uses the Variational Quantum Eigensolver (VQE) algorithm to estimate the ground-state (minimum) energy of a peptide-drug interaction. VQE is a hybrid quantum-classical algorithm: a parameterized quantum circuit (the ansatz) prepares a trial quantum state on a quantum processor or simulator, and a classical optimizer iteratively adjusts the circuit parameters to minimize the measured energy expectation value. The minimum energy found corresponds to the most stable configuration — i.e., the real binding energy of the drug-protein complex.

For hackathon execution, the quantum circuit is simulated using Qiskit Aer's statevector simulator on a classical CPU. The peptide fragment is limited to 5–10 amino acids of the target protein's binding pocket — specifically the HP (Hydrophobic-Polar) lattice model representation. This is the standard approach used in peer-reviewed quantum drug discovery literature, including IBM Research's published work on protein folding with quantum computers. The framing is always "quantum-ready": the architecture is built for real QPU execution; the demo runs on simulation.

### 5.1.2 Functional Requirements

- The system SHALL accept a drug molecule SMILES string and a target protein PDB sequence as input.
- The system SHALL encode the target peptide fragment as an HP lattice model with up to 10 residues.
- The system SHALL map the HP model to a QUBO (Quadratic Unconstrained Binary Optimization) problem.
- The system SHALL construct a parameterized quantum circuit (RealAmplitudes or EfficientSU2 ansatz, depth 2) for VQE optimization.

- The system SHALL use COBYLA or SPSA classical optimizer with maximum 200 iterations.
- The system SHALL return a normalized binding affinity score (0.0 to 1.0, where 1.0 = strongest predicted binding).
- The system SHALL complete VQE computation within 30 seconds on standard laptop hardware using Aer simulator.
- The system SHALL display the energy convergence plot as part of the output dashboard.

### 5.1.3 Acceptance Criteria

7. Given a valproic acid SMILES input and Nav1.2 peptide fragment, when the VQE runs, then it SHALL converge to a binding score between 0.65 and 0.90 (therapeutic range, calibrated against published DFT values).
8. Given lamotrigine SMILES and same Nav1.2 fragment, when the VQE runs, then lamotrigine SHALL produce a binding score within ±0.05 of valproic acid (both are effective against Nav1.2 — the difference appears in off-target scanning).
9. Given any SMILES string, when VQE fails to converge in 200 iterations, then the system SHALL flag "Low Confidence Score" and proceed with the last best estimate rather than crashing.

## F-002 — Off-Target Protein Panel Scanning

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|---|---|
| **Feature ID** | F-002 |
| **Priority** | P0 — Critical Core Feature |
| **Layer** | Layer 3 — Safety Screening |
| **Technology** | DeepChem (GraphConv model), ChEMBL database, RDKit |
| **Data Source** | ChEMBL 33 (2.4M+ bioactivity data points), SIDER database, FDA FAERS |

### 5.2.1 Scientific Description

Off-target binding is the primary mechanism of drug side effects. When a drug molecule is designed to bind Protein A (the intended target), its molecular shape and electrostatics may also interact with Protein B, C, or D — proteins the drug was never meant to touch. These unintended interactions produce unwanted biological effects that manifest as side effects in patients.

The off-target panel in QuantaMed contains 8 clinically critical proteins selected for their known association with serious adverse drug reactions. The panel was curated from FDA adverse

event data (FAERS) and the SIDER side effect database. For each candidate drug, a DeepChem GraphConv model predicts the probability of binding affinity above 10μM (the conventional threshold for clinically significant binding) at each panel protein.

### 5.2.2 Off-Target Panel Definition

| Panel Protein | Biological Role | Clinical Risk if Bound | Flagged For |
|---|---|---|---|
| hERG (KCNH2) | Cardiac potassium channel | QT interval prolongation → arrhythmia | All candidates |
| Androgen Receptor (AR) | Androgen signaling | PCOS, masculinization (females) | Female patients |
| CYP1A2 | Hepatic drug metabolism | Drug-drug interactions via enzyme inhibition | All candidates |
| CYP2D6 | Hepatic drug metabolism | Accumulation in PM phenotype patients | CYP2D6 PM patients |
| CYP3A4 | Major hepatic metabolizer (50% of drugs) | Broad DDI risk | All candidates |
| CYP2C9 | Warfarin/NSAID metabolism | Bleeding risk in co-medicated patients | Patients on warfarin/NSAIDs |
| Muscarinic M1 Receptor | Cholinergic signaling | Cognitive impairment, dry mouth | Elderly patients |
| Histamine H1 Receptor | Immune/sleep regulation | Sedation, weight gain | All candidates |

### 5.2.3 Functional Requirements

- The system SHALL score each candidate drug against all 8 panel proteins using DeepChem GraphConv or a pre-trained QSAR model.
- The system SHALL produce a risk matrix with predicted binding probability (0.0–1.0) for each drug-protein pair.
- Scores above 0.6 SHALL trigger a HIGH off-target risk flag. Scores between 0.4–0.6 SHALL trigger MODERATE. Below 0.4 SHALL be LOW.
- For female patients, the Androgen Receptor score SHALL be weighted 2× in the overall safety score calculation.
- The system SHALL generate a human-readable explanation for each HIGH flag, specifying the mechanism and the predicted clinical consequence.

## F-003 — Pharmacogenomics & CYP Compatibility Engine

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| **Feature ID** | F-003 |
| **Priority** | P0 — Critical Core Feature |
| **Layer** | Layer 4 — Personalized Compatibility |
| **Technology** | PharmVar database lookup, custom Python risk modifier engine |
| **Standards** | CPIC (Clinical Pharmacogenomics Implementation Consortium) guidelines |

### 5.3.1 Scientific Description

The CYP450 enzyme system is responsible for the phase I metabolism of approximately 75% of all marketed drugs. Genetic variants (alleles) in the genes encoding these enzymes determine how quickly a patient metabolizes specific drugs. QuantaMed uses a patient's CYP phenotype to apply individualized risk multipliers to both efficacy predictions and adverse event probabilities.

The system implements CPIC guideline recommendations — the gold standard in clinical pharmacogenomics — to translate genotype-to-phenotype mappings into actionable dosing and drug selection guidance. CPIC guidelines are published by a consortium of academic medical centers and have been endorsed by the FDA and multiple international regulatory bodies.

### 5.3.2 CYP Phenotype Risk Multiplier Table

| CYP Enzyme | Poor Metabolizer Effect | Ultrarapid Metabolizer Effect | Key Drugs Affected |
|---|---|---|---|
| CYP2D6 | Drug accumulation (3–5× normal) | Drug under-exposure (inadequate levels) | Codeine, tramadol, many antidepressants, risperidone |
| CYP2C19 | Proton pump inhibitor accumulation; clopidogrel non-activation | Rapid acid return; clopidogrel over-activation | Omeprazole, clopidogrel, citalopram, escitalopram |
| CYP2C9 | Warfarin accumulation (bleeding risk); NSAID toxicity | Reduced NSAID/warfarin levels | Warfarin, phenytoin, losartan, valproic acid (partial) |
| CYP3A4 | Statin myopathy risk; immunosuppressant toxicity | Reduced statin/immunosuppressant efficacy | Statins, tacrolimus, cyclosporine, many antiepileptics |

| | | | |
|---|---|---|---|
| CYP1A2 | Clozapine/olanzapine accumulation | Reduced antipsychotic levels | Clozapine, olanzapine, theophylline, caffeine |

### 5.3.3 Functional Requirements

- The system SHALL accept CYP2D6, CYP2C19, CYP2C9, CYP3A4, and CYP1A2 phenotype as input fields (Poor Metabolizer / Intermediate / Normal / Ultrarapid).
- The system SHALL apply CPIC-defined risk multipliers to each drug candidate's efficacy and safety scores.
- For Gabi (CYP2C9 IM): valproic acid's safety score SHALL be penalized by 1.4× the normal accumulation factor.
- The system SHALL generate a CYP interaction alert if a candidate drug is a substrate, inhibitor, or inducer of a CYP enzyme for which the patient has a non-normal phenotype.
- Alert format SHALL follow: "[Drug] is a [substrate/inhibitor/inducer] of CYP[X]. Patient's [phenotype] status increases [risk description] by approximately [X]×."

## F-004 — Dose-Adjusted Drug-Drug Interaction (DDI) Engine

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|---|---|
| **Feature ID** | F-004 |
| **Priority** | P0 — Critical Core Feature |
| **Layer** | Layer 4 — Personalized Compatibility |
| **Technology** | One-compartment PK model (Python), DDI rule database |
| **Scientific Model** | One-compartment pharmacokinetic model: C(t) = (F × Dose / Vd) × e^(-ke × t) |

### 5.4.1 Scientific Description

Pharmacokinetics (PK) is the study of what the body does to a drug — how it is absorbed, distributed, metabolized, and eliminated. The simplest PK model, the one-compartment model, treats the body as a single well-mixed compartment. Drug concentration at time t is given by: C(t) = (F × Dose / Vd) × e^(-ke × t), where F is bioavailability, Vd is volume of distribution, and ke is the elimination rate constant.

Drug-drug interactions at CYP enzymes can dramatically alter ke — an inhibitor drug reduces ke for the substrate drug, causing accumulation. QuantaMed calculates predicted steady-state plasma concentration for each candidate drug at the patient's prescribed dose, then applies DDI severity scaling based on predicted concentration — not just a binary interaction flag.

### 5.4.2 Functional Requirements

- The system SHALL compute predicted steady-state plasma concentration (Css) for each drug candidate using published PK parameters (F, Vd, ke, t1/2) from DrugBank database.
- Css calculation SHALL incorporate the patient's CYP phenotype as a modifier to ke (e.g., CYP2C9 IM reduces ke by 30% for CYP2C9 substrates).
- DDI severity SHALL be displayed as CRITICAL / HIGH / MODERATE / LOW based on Css thresholds rather than binary flags.
- The system SHALL display: "At [dose]mg/day, predicted Css = [X] µg/mL. Therapeutic range: [Y–Z]. CYP2C9 IM status elevates Css by approximately [N]%."
- For Gabi: at 1000mg/day valproic acid with CYP2C9 IM status, the system SHALL flag Css at or above 78 µg/mL as ELEVATED and note proximity to hepatotoxicity threshold of 100 µg/mL.

## F-005 — ADMET Prediction & Blood-Brain Barrier Scoring

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|---|---|
| **Feature ID** | F-005 |
| **Priority** | P0 — Critical Core Feature |
| **Layer** | Layer 3 — Safety Screening |
| **Technology** | DeepChem pre-trained ADMET models, RDKit molecular descriptors |
| **ADMET Definition** | Absorption, Distribution, Metabolism, Excretion, Toxicity |

### 5.5.1 Why ADMET Matters

A drug candidate can have perfect binding affinity for its target protein and still fail completely as a medicine if: (a) it cannot be absorbed from the gut after oral administration, (b) it cannot cross the blood-brain barrier for CNS conditions, (c) it is metabolized too quickly to maintain effective plasma levels, (d) it is eliminated so slowly that it accumulates to toxic concentrations, or (e) it is directly toxic to the liver, kidneys, or cardiac tissue. ADMET prediction identifies these failure modes computationally before any synthesis or animal testing.

For neurological and psychiatric drugs specifically, blood-brain barrier (BBB) penetration is the single most critical ADMET property. The BBB is a selective membrane formed by tight junctions between endothelial cells in brain capillaries. Many drugs that work perfectly in a test tube cannot cross this barrier and therefore have zero CNS efficacy. A drug candidate for epilepsy that cannot penetrate the BBB is useless, regardless of its Nav1.2 binding affinity.

### 5.5.2 Functional Requirements

- The system SHALL predict BBB penetration probability (0.0–1.0) for each candidate using DeepChem's pre-trained BBB model (trained on 2053 compounds from the BBB dataset).
- Candidates with BBB score below 0.50 SHALL be flagged "BBB PENETRATION: LOW — CNS efficacy unlikely" and penalized heavily in overall ranking for neurological/psychiatric indications.
- The system SHALL predict aqueous solubility (logS), lipophilicity (logP), plasma protein binding (%PPB), and half-life (t1/2) using RDKit-calculated molecular descriptors.
- The system SHALL predict hepatotoxicity probability using DeepChem's Tox21 model output on the "SR-HSE" (hepatic stress response element) assay.
- All ADMET scores SHALL be displayed in a normalized radar chart for visual comparison across candidates.

## F-006 — Lipinski Rule-of-Five Manufacturability Scoring

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|---|---|
| **Feature ID** | F-006 |
| **Priority** | P1 — High Priority |
| **Layer** | Layer 3 — Manufacturability Assessment |
| **Technology** | RDKit (molecular descriptor calculation) |
| **Scientific Basis** | Lipinski's Rule of Five (Ro5) — published 1997, Pfizer |

### 5.6.1 Scientific Description

Lipinski's Rule of Five is a set of five molecular property criteria that predict whether a drug candidate is likely to be orally bioavailable — a prerequisite for most practical drug formulations. The rules, derived empirically from analysis of thousands of marketed drugs, specify that an orally active drug should generally not violate more than one of the following: molecular weight ≤ 500 Da, logP (lipophilicity) ≤ 5, hydrogen bond donors ≤ 5, hydrogen bond acceptors ≤ 10, and molar refractivity between 40 and 130.

QuantaMed incorporates Ro5 as its manufacturability proxy — the hackathon theme explicitly requires manufacturability scoring, and Ro5 is the scientifically accepted, computationally accessible first-pass filter used by every major pharmaceutical company. Candidates that violate 2 or more rules are flagged as "Challenging to Manufacture/Formulate" and penalized in the overall ranking.

### 5.6.2 Functional Requirements

- The system SHALL calculate MW, logP, HBD, HBA, and TPSA for each candidate drug using RDKit from the SMILES string.

- Each property SHALL be displayed with a PASS/FAIL indicator relative to Lipinski thresholds.
- Manufacturability score SHALL be: 1.0 (0 violations) / 0.8 (1 violation) / 0.5 (2 violations) / 0.2 (3+ violations).
- Display SHALL include: "Valproic acid: MW 144 Da ✓, logP 2.75 ✓, HBD 1 ✓, HBA 2 ✓ — Manufacturability: EXCELLENT (0 violations)."

## F-007 — TRIBE v2 Symptom-State Visualizer

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|---|---|
| **Feature ID** | F-007 |
| **Priority** | P0 — Critical Demo Feature |
| **Layer** | Layer 1 (Patient Modeling) & Layer 6 (Visualization) |
| **Technology** | TRIBE v2 (Meta AI Demo Platform — aidemos.atmeta.com/tribev2) |
| **Role in System** | Primary visual storytelling tool — NOT a drug simulator |

### 5.7.1 Role Clarification

TRIBE v2 is NOT used to simulate drug pharmacology. It is used as a patient symptom-state visualizer — mapping the patient's clinical condition (symptom burden, medication effects, behavioral state) to sensory and neural visual metaphors. The science of drug-protein interaction happens in the quantum and ADMET layers. TRIBE v2 makes that science human and emotional for judges, clinicians, and patients.

### 5.7.2 Mapping Strategy — Condition to Sensory Input

| Patient State | Clinical Correlate | TRIBE v2 Sensory Mapping | Visual Output Goal |
|---|---|---|---|
| Pre-treatment (seizing) | High seizure frequency, high stress | Dissonant audio, chaotic visual stimuli | Chaotic, overwhelming brain activity visualization |
| On valproic acid (partial control) | Reduced seizures but PCOS emerging | Calmer audio but disrupted hormonal rhythms | Partially stabilized but metabolically stressed state |
| On lamotrigine (optimized) | Full seizure control, clean side effect profile | Nature sounds, harmonic visual input, stable rhythm | Calm, integrated, normalized brain state visualization |

| | | | |
|---|---|---|---|
| Side effect state (PCOS) | Weight gain, hormonal dysregulation, mood effects | Discordant low-frequency tones, slow rhythm disruption | Reward pathway dysregulation — dimmed nucleus accumbens |

### 5.7.3 Functional Requirements

- The system SHALL display two TRIBE v2 side-by-side simulations: Gabi on valproic acid vs. Gabi on lamotrigine.
- Each simulation SHALL map clinical state variables (seizure score, side effect burden, adherence) to specific sensory input parameters in TRIBE v2.
- The visualization SHALL include narrative labels: "Brain State: Partial Seizure Control — Hormonal Stress Detected" vs. "Brain State: Full Seizure Control — Metabolically Stable."
- TRIBE v2 output SHALL be embedded as an iframe or recorded screen capture in the demo dashboard.

## F-008 — Reinforcement Learning Treatment Decision Engine

| | |
|---|---|
| **Feature ID** | F-008 |
| **Priority** | P1 — High Priority |
| **Layer** | Layer 5 — RL Decision Engine |
| **Technology** | Python, NumPy — rule-based synthetic RL environment (no external RL library required) |
| **RL Type** | Tabular Q-learning on discrete synthetic state space |

### 5.8.1 Scientific Description

The RL engine models the treatment decision problem as a Markov Decision Process (MDP). The state space encodes the patient's current clinical status. The action space contains three decisions: CONTINUE current therapy, SWITCH to next-ranked candidate, or REDUCE dose. The reward function balances therapeutic benefit (seizure reduction) against risk burden (side effect severity), with a temporal discount factor that models how risk accumulates over months.

### 5.8.2 State Space Definition

| State Variable | Range | Source | Clinical Meaning |
|---|---|---|---|

| | | | |
|---|---|---|---|
| seizure_frequency | 0–10 per month | Patient report / EEG summary | Primary efficacy measure |
| side_effect_burden | 0.0–1.0 | ADMET score + CYP modifier | Weighted composite of predicted adverse effects |
| adherence_rate | 0.0–1.0 | Simulated (default 0.85) | Proportion of doses taken on schedule |
| time_on_therapy | 0–24 months | Patient profile | Time-dependent risk accumulation factor |
| hormonal_risk | 0.0–1.0 | Off-target AR binding score × gender modifier | Endocrine disruption risk (female-specific) |
| hepatic_stress | 0.0–1.0 | Css / hepatotoxicity threshold | Liver burden from drug accumulation |

### 5.8.3 Reward Function

The reward at each timestep is computed as: R = w1 × efficacy_improvement − w2 × side_effect_increase − w3 × adherence_penalty − w4 × temporal_risk_accumulation, where the weights w1=0.4, w2=0.3, w3=0.15, w4=0.15 reflect the relative clinical importance of each factor. The temporal_risk_accumulation term increases linearly with time_on_therapy for drugs with known chronic toxicity profiles (like valproic acid's androgenic effects).

### 5.8.4 Functional Requirements

- The RL engine SHALL simulate 12 months of therapy in discrete monthly timesteps.
- At each timestep, the engine SHALL evaluate the current state and apply the policy to select an action.
- The engine SHALL output a trajectory plot: seizure frequency, side effect burden, and cumulative reward over 12 months for each candidate drug.
- For Gabi's case: valproic acid SHALL produce declining cumulative reward after month 6 (PCOS emerging). Lamotrigine SHALL produce rising or stable cumulative reward across 12 months.
- The engine SHALL display its recommendation in plain language: "SWITCH RECOMMENDED: Valproic acid cumulative risk score exceeds threshold at month 6. Lamotrigine maintains superior risk-benefit ratio throughout the simulation period."

## F-009 — Drug Comparison Matrix

| | |
|---|---|
| **Feature ID** | F-009 |
| **Priority** | P1 — High Priority |
| **Layer** | Layer 6 — Visualization |
| **Technology** | React / Streamlit / HTML dashboard |
| **Purpose** | Side-by-side visual comparison of all candidate drugs across all scoring dimensions |

### 5.9.1 Matrix Structure

| Scoring Dimension | Source Feature | Display Format | Weight in Final Score |
|---|---|---|---|
| Quantum Binding Affinity | F-001 VQE output | Score 0–100 + color bar | 25% |
| Overall Safety Score | F-002 off-target panel | Score 0–100 + risk badge | 25% |
| Patient Genomic Compatibility | F-003 CYP engine | Multiplier + adjusted score | 20% |
| ADMET Profile | F-005 DeepChem | Radar chart + composite score | 15% |
| BBB Penetration | F-005 DeepChem | Percentage + PASS/FAIL | 10% |
| Manufacturability (Ro5) | F-006 RDKit | Violation count + score | 5% |

### 5.9.2 Functional Requirements

- Matrix SHALL display all candidate drugs as columns and all scoring dimensions as rows.
- Best-performing drug in each row SHALL be highlighted in green. Worst SHALL be highlighted in amber or red.
- Final composite score SHALL appear at the bottom of each column with a clear RECOMMENDED / ACCEPTABLE / NOT RECOMMENDED badge.
- Matrix SHALL be filterable: user can exclude candidates that fail any single criterion (e.g., "Hide all candidates with BBB < 50%").
- For Gabi's demo: lamotrigine SHALL score highest overall despite similar binding affinity to valproic acid, because of its significantly superior off-target and ADMET profiles.

## F-010 — Risk Timeline Visualization

| | |
|---|---|
| **Feature ID** | F-010 |
| **Priority** | P1 — High Priority |
| **Layer** | Layer 6 — Visualization |
| **Technology** | Plotly (Python) or Chart.js |
| **Data Source** | RL engine trajectory output (F-008) |

The risk timeline graph plots predicted interaction risk and clinical outcome trajectory over a 12-month period for each candidate drug, conditioned on the patient's profile. For Gabi, this graph directly shows the divergence point — the moment around month 4–6 when valproic acid's cumulative hormonal risk score crosses above the safety threshold while lamotrigine's remains flat. This single graph is arguably the most impactful single visualization in the entire demo, because it makes the temporal dimension of drug safety visible in a way that no existing clinical tool can.

- The graph SHALL display: cumulative risk score (Y-axis) over months 1–12 (X-axis) for each candidate drug as a separate line.
- The graph SHALL include threshold lines: "SAFETY THRESHOLD" (horizontal dashed line) and "PCOS EMERGENCE WINDOW" (shaded region, months 3–6).
- The graph SHALL display an annotation: "Valproic acid crosses risk threshold at month [X]. Lamotrigine remains below threshold throughout simulation."
- Adherence scenarios SHALL be toggle-able: 100% adherence vs. 70% adherence (showing how missed doses affect the risk trajectory differently for each drug).

## F-011 — Population Benchmark Comparison

| | |
|---|---|
| **Feature ID** | F-011 |
| **Priority** | P2 — Medium Priority |
| **Layer** | Layer 6 — Visualization |
| **Data Source** | FDA FAERS database (pre-loaded statistics for demo drugs), SIDER |

Population benchmarking contextualizes an individual patient's predicted risk against the population of patients who have actually taken this drug. For Gabi, this means displaying: "Predicted weight gain risk: 2.1× higher than population average for females under 30 on valproic acid (source: FAERS, 12,847 reports)." This transforms abstract model scores into clinically interpretable comparisons that clinicians and patients can immediately understand.

- Pre-loaded FAERS statistics SHALL be available for valproic acid and lamotrigine across the following adverse event categories: weight gain, hormonal disruption, hepatic events, CNS effects, teratogenicity.
- Each risk statistic SHALL display: absolute rate in population, relative rate vs. average, confidence interval, and data volume (number of reports).
- Female-specific risk statistics SHALL be displayed prominently for female patients with a "⚠ Sex-Specific Risk" label.

## F-012 — Exportable Clinical PDF Report

| | |
|---|---|
| **Feature ID** | F-012 |
| **Priority** | P1 — High Priority |
| **Layer** | Layer 6 — Reporting |
| **Technology** | ReportLab (Python PDF generation) |
| **Output** | One-page clinical summary formatted for handoff to physician |

The PDF report is the artifact that bridges the computational world of QuantaMed to the real clinical world. It is formatted as a structured clinical document that a neurologist can bring to a treatment planning meeting. It contains: patient profile summary, top 3 ranked drug candidates with composite scores, key risk flags in plain language, CYP interaction alerts, TRIBE v2 brain state thumbnail, and QR code linking to the full interactive dashboard.

- PDF SHALL be generated in under 5 seconds from dashboard output.
- PDF SHALL include QuantaMed branding, patient ID, date, and a "Quantum-Enhanced Analysis" badge.
- ALL drug scores SHALL include a disclaimer: "This analysis is a computational decision support tool. Clinical judgment of a licensed physician is required for all prescribing decisions."

# 6. Complete Technical Stack

## 6.1 Backend / Computational Layer

| Component | Library/Tool | Version | Role | Install Command |
|---|---|---|---|---|
| Quantum Simulation | Qiskit + Qiskit-Nature | 0.45+ | VQE protein binding simulation | pip install qiskit qiskit-nature |
| Quantum Simulator | Qiskit Aer | 0.13+ | Classical simulation of quantum circuits | pip install qiskit-aer |
| Molecular ML | DeepChem | 2.7+ | ADMET, BBB, off-target prediction | pip install deepchem |
| Cheminformatics | RDKit | 2023.03+ | Lipinski Ro5, SMILES parsing, fingerprints | pip install rdkit |
| PK Modeling | NumPy + SciPy | latest | One-compartment PK equation, ODE solving | pip install numpy scipy |
| RL Engine | Pure Python + NumPy | N/A | Tabular Q-learning on synthetic state space | No additional install |
| DDI Database | DrugBank API / Local | v5.1 | PK parameters, CYP substrate tables | API key + local cache |
| Bioactivity Data | ChEMBL (REST API) | 33 | Off-target binding IC50 data | pip install chembl-webresource-client |
| PDF Generation | ReportLab | 4.0+ | Clinical PDF report export | pip install reportlab |

## 6.2 Frontend / Visualization Layer

| Component | Technology | Role |
|---|---|---|
| Main Dashboard | Streamlit (Python) or React.js | Primary interactive interface for drug comparison, timeline, and report generation |
| Interactive Charts | Plotly (Python/JS) | Risk timeline graph, ADMET radar chart, VQE convergence plot |
| Molecular Viewer | py3Dmol (embedded in Streamlit) | Interactive 3D visualization of drug-protein binding pose |
| TRIBE v2 Integration | iframe embed (aidemos.atmeta.com/tribev2) | Patient symptom-state brain visualization (before/after comparison) |

| | | |
|---|---|---|
| Drug Comparison Matrix | Pandas + Streamlit AgGrid | Sortable, filterable side-by-side candidate comparison table |
| CYP Mechanism Diagram | SVG animation (custom) or Matplotlib | Visual explanation of CYP enzyme inhibition mechanism |

## 6.3 Data Sources

| Data Source | Type | Data Provided | Access Method |
|---|---|---|---|
| PubChem | Public API | SMILES strings, molecular properties for all FDA drugs | REST API (free, no key) |
| ChEMBL 33 | Public Database / API | 2.4M+ bioactivity measurements across drug-target pairs | REST API + local SQLite mirror |
| DrugBank | Licensed/Academic API | PK parameters (Vd, ke, F, t1/2), CYP substrate/inhibitor tables | Free academic registration |
| PharmVar | Public Database | CYP gene allele definitions and phenotype mappings | Direct download (free) |
| FDA FAERS | Public Database | Adverse event reports — population-level side effect rates | OpenFDA REST API (free) |
| SIDER | Public Database | Drug-side effect associations from drug labels | Direct download (free) |
| RCSB Protein Data Bank | Public Database | Protein 3D structures and binding pocket coordinates | REST API (free) |

# 7. Non-Functional Requirements

## 7.1 Performance

- VQE quantum simulation (5-residue peptide, Aer simulator): SHALL complete in ≤ 30 seconds on a standard laptop (Intel i7, 16GB RAM).
- Full pipeline execution (all 5 candidates, all scoring layers): SHALL complete in ≤ 3 minutes end-to-end.
- Dashboard page load: SHALL render in ≤ 2 seconds after pipeline completion.
- PDF report generation: SHALL complete in ≤ 5 seconds.
- TRIBE v2 visualization: SHALL load in ≤ 10 seconds (dependent on network connectivity to aidemos.atmeta.com).

## 7.2 Reliability

- If VQE fails to converge, system SHALL fall back to a classical Hartree-Fock approximation and display a "Classical Approximation Used" badge — the pipeline SHALL NOT halt.
- If DeepChem model prediction fails (model file missing), system SHALL use pre-computed lookup table for demo drugs and display a "Cached Prediction" badge.
- If external API (PubChem, ChEMBL) is unavailable, system SHALL use locally cached SMILES and bioactivity data for the 5 demo candidate drugs.
- The system SHALL maintain demo functionality with zero internet connectivity for all core features (using pre-cached data), with TRIBE v2 being the only internet-dependent component.

## 7.3 Usability (Hackathon Demo Context)

- The entire Gabi demo flow SHALL be completable in under 5 minutes from startup to PDF report generation.
- A judge with no pharmacology background SHALL be able to understand the Drug Comparison Matrix result at a glance.
- All technical scores SHALL have human-readable interpretations: never just a number, always a label (EXCELLENT / GOOD / MODERATE RISK / HIGH RISK).
- The TRIBE v2 before/after comparison SHALL be immediately emotionally intelligible without explanation.

## 7.4 Scientific Integrity

- Every scoring algorithm SHALL cite its source (published paper, database, or CPIC guideline) in the UI tooltip.
- No scoring component SHALL claim capabilities beyond what is computationally demonstrated in the demo.
- The quantum module SHALL be explicitly labeled "Quantum-Ready Module — Simulated on Classical Hardware via Qiskit Aer."
- The PDF report SHALL include a mandatory disclaimer: "For research and decision support purposes only. Not for clinical use without physician review."

# 8. End-to-End Demo Flow — Gabi's Journey

This section defines the exact sequence of user interactions and system outputs for the primary hackathon demonstration. Every team member must know this flow by heart. This is the story we are telling.

## Step 1 — Patient Profile Input (30 seconds)

The demo operator opens the QuantaMed dashboard. A "New Patient" panel is visible on the left side. The operator enters Gabi's information: Age 24, Female, Diagnosis: Juvenile Myoclonic Epilepsy, Current Drug: Valproic Acid 1000mg/day (9 months), CYP2D6 Poor Metabolizer, CYP2C9 Intermediate Metabolizer, Hepatic Function: Mildly Elevated ALT. A patient agent card appears in the TRIBE v2 panel, labeled "Gabi — Pre-Optimized Treatment." The agent shows chaotic sensory stimulation patterns.

## Step 2 — Quantum Binding Simulation (60–90 seconds)

The operator clicks "Run Quantum Analysis." A progress indicator shows: "Encoding Nav1.2 peptide fragment as HP lattice model... Building VQE ansatz circuit... Optimizing with COBYLA... Convergence achieved." Two graphs appear: the VQE energy convergence plot (showing the optimizer finding the minimum), and the binding affinity scores for all 5 candidates as a bar chart. Valproic acid and lamotrigine both show strong binding (0.82 and 0.79 respectively). The operator notes: "They're equally good at targeting the right protein. The difference is what they do to everything else."

## Step 3 — Off-Target Safety Scan (30 seconds)

The off-target risk matrix populates automatically. Valproic acid's Androgen Receptor column lights up RED (binding probability 0.71). A modal alert appears: "HIGH OFF-TARGET BINDING DETECTED: Valproic acid shows predicted androgen receptor binding. For female patients under 30, this is associated with 2.8× elevated PCOS risk. Risk accumulates over 3–6 months of chronic exposure." Lamotrigine's entire row is green. The operator says: "This is the flag that Gabi's prescribing physician never had."

## Step 4 — CYP Genomic Compatibility (20 seconds)

The genomic compatibility panel shows Gabi's CYP phenotype. A warning appears under valproic acid: "CYP2C9 Intermediate Metabolizer: Predicted Css = 78.4 µg/mL at 1000mg/day — 31% above normal accumulation. Hepatotoxicity threshold: 100 µg/mL. Current trajectory: CAUTION." Lamotrigine shows: "Not a primary CYP2C9 substrate — genomic risk: LOW."

## Step 5 — Drug Comparison Matrix (20 seconds)

The full comparison matrix renders. Five drugs as columns, six scoring dimensions as rows. Lamotrigine has the highest composite score (84/100). Valproic acid scores 61/100 — competitive on efficacy but penalized heavily on safety and genomic compatibility. A green "RECOMMENDED" badge appears under lamotrigine. A red "NOT RECOMMENDED FOR THIS PATIENT" badge appears under valproic acid. The operator says: "Gabi has been on the second-best option for nine months. We could have told her this on day one."

## Step 6 — Risk Timeline & RL Engine (30 seconds)

The risk timeline graph animates. Two lines grow across 12 months. Valproic acid's risk score crosses the safety threshold at month 5 — a red shaded zone labeled "PCOS EMERGENCE WINDOW" highlights this crossing. Lamotrigine's line stays flat and green. The RL engine displays its verdict: "SWITCH RECOMMENDED. Cumulative risk-benefit ratio favors lamotrigine at all timepoints for this patient profile. Estimated efficacy equivalence: 94%. Estimated side effect reduction: 67%."

## Step 7 — TRIBE v2 Before/After Visualization (30 seconds)

Two TRIBE v2 frames appear side by side. Left panel, labeled "Gabi — Valproic Acid (Current)": disrupted sensory patterns, amber and red tones, discordant audio cues if sound is on, label "Reward Pathway: Dysregulated — Hormonal Stress Active." Right panel, labeled "Gabi — Lamotrigine (Recommended)": calm nature sounds, green and blue tones, smooth rhythmic patterns, label "Reward Pathway: Normalized — Metabolically Stable." The room goes quiet for two seconds. That's the moment.

## Step 8 — PDF Report Generation (10 seconds)

The operator clicks "Export Clinical Report." A PDF downloads in 4 seconds. It contains Gabi's full profile, the top 3 ranked drugs with composite scores, the three critical risk flags in plain

language, and a recommendation statement. The bottom carries the disclaimer. The operator says: "This is what Gabi's neurologist would receive in their inbox. It is built from quantum simulation, population genomics, pharmacokinetics, and 12 months of simulated therapy — delivered in under five minutes."

# 9. The Quantum Argument — Why VQE Here, Why Now

Hackathon judges will ask: "Why quantum? Couldn't you do this classically?" This section provides the complete, technically honest answer.

## 9.1 The Honest Answer

For the peptide sizes simulated in this demo (5–10 amino acids), classical computers can simulate the quantum calculation with acceptable accuracy. We are not claiming quantum advantage at this scale. What we are claiming — and what is scientifically true — is that the VQE architecture we have built is the same architecture that will achieve quantum advantage at scale, when run on real quantum hardware with fault-tolerant qubits.

The threshold for quantum advantage in molecular simulation is generally considered to be 50–100+ electrons — corresponding to peptide chains of 15–25 amino acids or small drug molecules in complex with their binding pocket. Current quantum hardware (IBM Eagle: 127 qubits, IBM Osprey: 433 qubits) is approaching the noise threshold required to demonstrate this.

Our system is "quantum-ready" in the most meaningful sense: the algorithm, the problem encoding, and the output interpretation are all quantum-native. The hardware is the only limitation, and it is a temporary one.

## 9.2 Why This Framing is Scientifically Legitimate

Every major pharmaceutical company — Pfizer, Roche, Biogen, AstraZeneca — has active quantum computing research programs specifically for molecular simulation. IBM, Google, and IonQ have published peer-reviewed papers showing VQE applied to molecular systems. The FDA has published guidance on quantum computing applications in drug development.

QuantaMed is building on a scientifically established foundation, not a speculative one.

In a hackathon context, this framing is not only legitimate — it is the correct framing. The goal is to demonstrate a quantum-native architecture that solves a real problem. We are doing exactly that.

# 10. Release Criteria & Success Metrics

## 10.1 Hackathon Demo Success Criteria

| Criterion | Measurement | Target | Critical? |
|---|---|---|---|
| End-to-End Demo Runs | Complete Gabi flow from input to PDF without crash | 3 consecutive successful runs | YES |
| VQE Convergence | Energy converges to minimum within iteration limit | ≤ 200 iterations for 5-residue peptide | YES |
| Lamotrigine Ranks #1 | Composite score positions lamotrigine as top candidate | Score ≥ 10 points above valproic acid | YES |
| PCOS Risk Flag Active | Androgen receptor off-target binding flag fires for valproic acid | Binding probability > 0.60 for AR | YES |
| TRIBE v2 Loads | Both patient states visible in TRIBE v2 | Both panels load within 10 seconds | YES |
| PDF Exports | Clinical report PDF generates and opens | Within 5 seconds, no formatting errors | YES |
| Demo Duration | Full story told within time limit | Under 5 minutes for standard demo | YES |
| Drug Comparison Matrix | All 5 candidates visible with scores | Color coding correct, badges accurate | HIGH |
| Risk Timeline Graph | Valproic acid crosses threshold at month 5–7 | Correct visual divergence point shown | HIGH |

## 10.2 Judge Evaluation Criteria Alignment

| Judging Criterion (Typical Hackathon) | How QuantaMed Addresses It |
|---|---|
| Innovation / Novelty | Quantum VQE binding simulation integrated with clinical DDI scoring — no existing tool does this combination |
| Technical Depth | Six interconnected simulation layers, each with published scientific basis and quantitative outputs |
| Problem-Solution Fit | Direct alignment with hackathon theme: AI to propose/prioritize drug candidates by efficacy, safety, manufacturability |
| Real-World Impact | Valproic acid-PCOS is a documented clinical tragedy affecting thousands of women — system would have prevented Gabi's harm |

| | |
|---|---|
| Presentation Quality | TRIBE v2 before/after is emotionally resonant; Drug Comparison Matrix is immediately intelligible; PDF is tangible deliverable |
| Completeness of Demo | Full end-to-end flow with real computation, real data, and a real patient story |
| Scalability Vision | Architecture is explicitly quantum-ready and designed for real QPU execution; business case for 12-year drug pipeline compression is articulated |

# 11. Development Timeline & Build Order

## 11.1 Recommended Build Sequence

The following sequence is optimized for hackathon execution: build the things that work first, add complexity in layers, and ensure the demo is demoable at every stage (so a crash never leaves you with nothing to show).

| Phase | Duration | Tasks | Demoable After? |
|---|---|---|---|
| Phase 0 — Environment | 1 hour | Install Qiskit, DeepChem, RDKit, Streamlit. Verify all imports succeed. Cache SMILES strings for 5 demo drugs. Cache pre-computed scores as fallback. | NO |
| Phase 1 — Quantum Core | 3 hours | Build VQE protein folding module. HP lattice encoding for 5-residue Nav1.2 fragment. Run for valproic acid and lamotrigine. Display convergence plot. | YES (quantum module only) |
| Phase 2 — Safety Layers | 2 hours | Build off-target panel scanner using pre-computed ChEMBL data. Implement Lipinski Ro5 scoring with RDKit. Implement BBB prediction with DeepChem. | YES (quantum + safety) |
| Phase 3 — Patient Engine | 2 hours | Build CYP phenotype modifier. Build one-compartment PK model for dose-adjusted DDI. Wire patient profile inputs. | YES (quantum + safety + patient) |
| Phase 4 — RL & Timeline | 2 hours | Build synthetic RL state machine. Generate 12-month trajectory plots with Plotly. Implement recommendation output. | YES (full backend) |
| Phase 5 — Dashboard UI | 2 hours | Build Streamlit or React dashboard. Drug comparison matrix with color coding. Connect all modules. Add TRIBE v2 iframes. | YES (full demo) |
| Phase 6 — PDF & Polish | 1 hour | Implement ReportLab PDF export. Add disclaimers, branding, QR code. Final end-to-end test × 3. | YES (complete) |

## 11.2 Critical Path Dependencies

- Phase 1 (VQE) must be complete before Phase 3 (patient engine) can wire VQE output as input.
- Phase 2 (safety layers) runs in parallel with Phase 3 — these are independent modules.
- Phase 4 (RL) depends on Phase 3 patient profile being finalized.
- Phase 5 (dashboard) requires all backend modules to have defined output schemas — even if computation is mocked.
- Phase 6 (PDF) is the last phase and has no dependencies other than Phase 5 being stable.

## 11.3 Risk Mitigation

**Risk: VQE takes too long to run live.** Mitigation: Pre-compute and cache VQE results for all 5 demo drugs. Display pre-computed results with the label "VQE Analysis Complete — Results Cached for Demo Speed." This is standard practice in demo contexts.

**Risk: DeepChem installation fails.** Mitigation: Pre-compute all ADMET and BBB scores for demo drugs and store as JSON lookup tables. The demo will use cached scores transparently.

**Risk: TRIBE v2 is down during demo.** Mitigation: Record a 60-second screen capture of the TRIBE v2 before/after comparison during preparation. Play the recording as a video if the live site is unavailable.

**Risk: Dashboard crashes mid-demo.** Mitigation: Maintain a PowerPoint backup with screenshots of every major output. Never demo without this backup prepared.

# 12. Appendix

## 12.1 Glossary

| Term | Definition |
|---|---|
| ADMET | Absorption, Distribution, Metabolism, Excretion, Toxicity — the five pharmacokinetic properties that determine whether a drug works safely in the human body |
| Ansatz | In VQE, a parameterized quantum circuit that approximates the target quantum state through variational optimization |
| BBB | Blood-Brain Barrier — selective membrane separating blood from brain tissue; must be penetrated for CNS drugs to work |
| ChEMBL | European database of bioactivity data for drug-like molecules (2.4M+ compounds) |
| COBYLA | Constrained Optimization BY Linear Approximations — classical optimizer used in VQE |
| CPIC | Clinical Pharmacogenomics Implementation Consortium — international organization publishing clinical guidelines for pharmacogenomics |
| CYP450 | Cytochrome P450 enzyme family — responsible for metabolizing ~75% of marketed drugs; genetic variants cause variable drug response |
| DDI | Drug-Drug Interaction — alteration of a drug's effect due to the presence of another drug |
| DeepChem | Open-source Python library for deep learning in drug discovery and molecular biology |
| FAERS | FDA Adverse Event Reporting System — database of adverse drug reactions reported to the FDA |
| GraphConv | Graph Convolutional Neural Network — ML architecture that operates on molecular graphs for property prediction |
| hERG | Human Ether-a-go-go Related Gene — cardiac potassium channel; off-target binding causes QT prolongation and arrhythmia |
| HP Model | Hydrophobic-Polar lattice model — simplified protein folding representation encoding residues as H (hydrophobic) or P (polar) |
| JME | Juvenile Myoclonic Epilepsy — common epilepsy syndrome; valproic acid and lamotrigine are both indicated treatments |
| Lipinski Ro5 | Rule of Five — five molecular property criteria predicting oral bioavailability of drug candidates |
| MDP | Markov Decision Process — mathematical framework for sequential decision-making; used in RL formulation |
| Nav1.2 | Voltage-gated sodium channel alpha subunit 1.2 — primary target of many antiepileptic drugs including valproic acid and lamotrigine |
| PCOS | Polycystic Ovarian Syndrome — hormonal disorder associated with valproic acid use in females; mechanism involves androgenic pathway disruption |

| | |
|---|---|
| PK | Pharmacokinetics — the study of how the body processes a drug (absorption, distribution, metabolism, excretion) |
| QUBO | Quadratic Unconstrained Binary Optimization — mathematical problem formulation amenable to quantum annealing and VQE |
| RDKit | Open-source cheminformatics library for molecular property calculation and drug-likeness scoring |
| SMILES | Simplified Molecular Input Line Entry System — text representation of a chemical structure |
| SNP | Single Nucleotide Polymorphism — single-letter DNA variation between individuals; CYP SNPs determine metabolizer phenotype |
| TRIBE v2 | Meta AI multi-agent simulation platform (aidemos.atmeta.com/tribev2); used in QuantaMed as symptom-state visualizer |
| VQE | Variational Quantum Eigensolver — hybrid quantum-classical algorithm for finding the minimum energy state of a molecular system |

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## 12.3 Boilerplate Requirements

- All patient data used in demonstrations is synthetic and anonymized. Real patient records are never stored or transmitted.

- All scientific claims are accompanied by citations to peer-reviewed literature or validated databases.
- The quantum module is labeled as a simulation using Qiskit Aer and does not claim real quantum hardware execution.
- All predictive outputs include uncertainty estimates or confidence labels where applicable.
- The clinical PDF report includes a mandatory disclaimer on every page that outputs are for research/decision-support purposes only.
- The system complies with GDPR Article 17 (right to erasure) — all patient profile data can be cleared from the session with one action.

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## QuantaMed — Built for Gabi. Designed for Everyone.

*Every year, thousands of patients are prescribed drugs that don't fit their biology. We built QuantaMed because no one should have to be Gabi.*

**Version 1.0 | Hackathon Edition | April 2026 | QuantaMed Team**